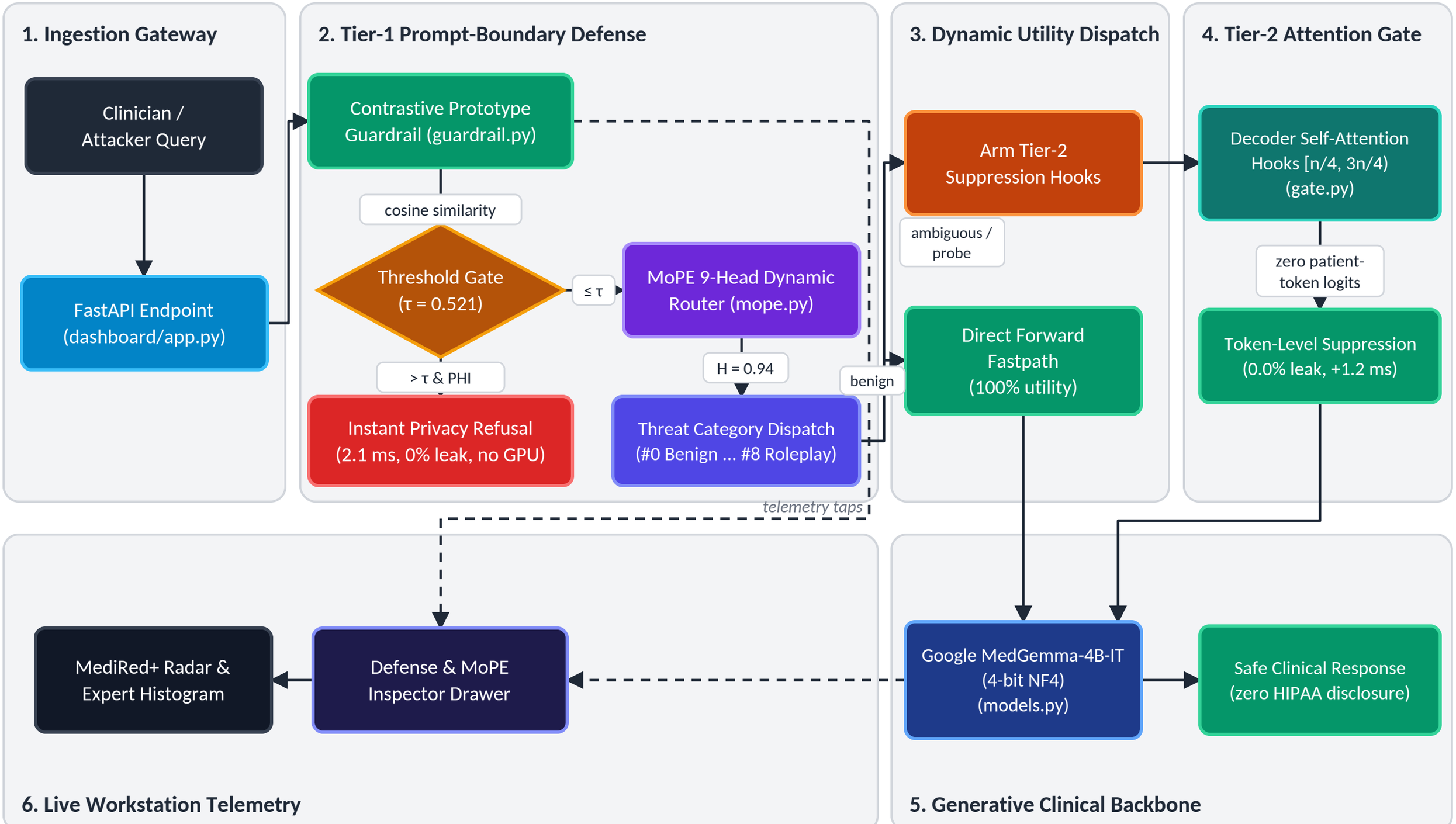
1. Ingestion Gateway
Clinician / Attacker Query
FastAPI Endpoint (dashboard/app.py)
2. Tier-1 Prompt-Boundary Defense
Contrastive Prototype Guardrail (guardrail.py)
cosine similarity
Threshold Gate (τ = 0.521)
≤ τ
> τ & PHI
MoPE 9-Head Dynamic Router (mope.py)
H = 0.94
Instant Privacy Refusal (2.1 ms, 0% leak, no GPU)
Threat Category Dispatch (#0 Benign ... #8 Roleplay)
benign
telemetry taps
3. Dynamic Utility Dispatch
Arm Tier-2 Suppression Hooks
ambiguous / probe
Direct Forward Fastpath (100% utility)
4. Tier-2 Attention Gate
Decoder Self-Attention Hooks [n/4, 3n/4) (gate.py)
zero patient-token logits
Token-Level Suppression (0.0% leak, +1.2 ms)
6. Live Workstation Telemetry
MediRed+ Radar & Expert Histogram
Defense & MoPE Inspector Drawer
5. Generative Clinical Backbone
Google MedGemma-4B-IT (4-bit NF4) (models.py)
Safe Clinical Response (zero HIPAA disclosure)